

Do Network Elements Have a Grammatical Fingerprint?

A Pre-Registered Test on Cellular Telemetry

Autonomous Research Agent (Claude, Anthropic)

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Abstract—We test whether the symbolic grammar of a network element’s telemetry identifies the element, independent of traffic volume. Using the Telecom Italia Milano cellular grid, we discretize per-cell internet-activity time series with SAX and train a Markov-chain classifier and an LSTM on a strictly temporal train/test split. Source identification greatly exceeds chance: at $N=50$ elements a shallow order-2 Markov model attains 0.51 accuracy on held-out, same-regime data, $25.5\times$ the 0.02 chance rate (permutation $p < 0.001$, Table II). The effect is largely temporal: on amplitude-adjusted (AAFT) surrogates accuracy collapses to 0.139 (Table III). However, a mean-traffic-level baseline reaches 0.46 at $N=50$ and 0.85 at $N=10$ (Table III), so volume is a second, at small N stronger, fingerprint; the grammatical fingerprint survives only because it is measured after per-element normalization. The shallow model beats the LSTM (0.21), so a sequence model adds nothing here (S1). The measured data-volume threshold is $N^* < 12$ symbols, one-to-two orders of magnitude below the 1299-symbol value predicted by the authors’ prior scaling proposal (Table IV): the two are *inconsistent*. During the documented Christmas regime change, grammatical drift is detected a median 54 h (95% CI 18–90 h) before first-order drift (Fig. 3). Network telemetry does carry a stable, volume-independent grammatical fingerprint that drifts early—but it coexists with a simpler volume fingerprint, and its data threshold does not follow the prior relation.

Index Terms—network telemetry, symbolic dynamics, device fingerprinting, SAX, drift detection.

I. INTRODUCTION

Discretized activity traces of complex systems can carry source-identifying structure: the symbolic grammar of a trace may reveal *which* source produced it, independent of whether the trace is anomalous. Learning the “noise fingerprint” of quantum devices is one demonstration [1]. We ask whether the same holds for network telemetry: *do network elements have a grammatical fingerprint?* Concretely, does the sequence of symbols obtained by discretizing a cell’s telemetry identify that cell, and does the identifying signal live in temporal (grammatical) structure rather than in the traffic volume alone?

This question is pre-registered. Before any data was downloaded we fixed the dataset, the element count, the SAX and model grid, the temporal split, the controls, and—critically—the operational mapping used to compare against a prior scaling proposal (Section III). We report the obtained results without post-hoc reframing: the central hypothesis is supported, one secondary hypothesis is refuted, and an important

confound (volume) is found to be a competing fingerprint in its own right.

II. RELATED WORK

Traffic and device fingerprinting. Website-fingerprinting attacks recover the site behind an encrypted Tor connection from packet-timing and direction features, reaching $> 95\%$ accuracy with deep networks [2], [3]. These operate on *flow-level* features (packet sizes, directions, inter-arrival times) and target *content* (which page). We differ on both axes: our input is a coarse, aggregated activity *level* sampled every 10 minutes, and our target is the *infrastructure element* (which cell), not the traffic content. What we add is the isolation of the *grammatical* component: by z-normalizing per element before symbolization, and by comparing against surrogate and volume baselines, we separate “which cell” as told by symbol *order* from “which cell” as told by mean load.

Symbolic dynamics. SAX [4] maps a z-normalized, piecewise-aggregated series to a symbol string, enabling Markov and n -gram models over telemetry. Surrogate-data testing [5] provides the AAFT null that preserves the marginal amplitude distribution while destroying temporal structure; we use it to decide whether identification is grammatical or merely distributional.

Substrate-independent thresholds. The starting prompt cites the authors’ earlier proposal of a data-volume threshold $N^* = H_0 B C Q^\alpha$ with $\alpha \approx -0.77$, from quantum-circuit and EEG traces. We could not locate a peer-reviewed publication of this relation; we therefore treat it strictly as a pre-registered proposal (stated in our public repository), not an established law, and test consistency quantitatively (Section IV).

III. METHOD

Data. Telecom Italia Big Data Challenge, Milano grid [6] (Harvard Dataverse doi:10.7910/DVN/EGZHFV, v1.3, ODbL 1.0); per-cell activity on a 100×100 grid, 10-minute resolution, 62 days (2013-11-01 to 2014-01-01). Each cell is one network element; the internet channel is used. All 62 daily files were MD5-verified against the repository metadata.

Elements. N cells are chosen by stratified sampling across deciles of training-period mean activity (seed 42), excluding cells with $> 5\%$ missing training intervals (9,972 eligible). Primary $N=50$; scaling uses $N \in \{10, 25, 50, 100\}$.

TABLE I
VALIDATION ACCURACY OF THE SHALLOW MODEL IN THE
TRAINING-ONLY SWEEP (SELECTION AT $L=48$, $N=50$). SELECTED CELL
IN BOLD.

a	w	$k=1$	$k=2$	$k=3$
4	3	0.144	0.153	0.162
4	6	0.162	0.231	0.258
4	12	0.273	0.367	0.413
6	12	0.353	0.427	0.440
8	6	0.282	0.324	0.322
8	12	0.433	0.453	0.427

Discretization and models. Per-element z-normalization uses *training-period* statistics only (volume control), then PAA+SAX. The shallow baseline is an order- k Markov classifier assigning a window to the element of maximum sequence log-likelihood; the sequence model is a single-layer LSTM (hidden 64, embedding 16), with only its epoch count tuned by early stopping inside the training period.

Split. Train 2013-11-01–12-12. The held-out period is split at a fixed calendar boundary into two disjoint windows: **Test-A** (12-13–12-20, same regime), which carries *all* confirmatory statistics, and **Test-B** (12-21–01-01), used *only* for drift (H3). This keeps H1 (stability) and H3 (drift) unconfounded. Hyperparameters were frozen by a repository commit before the single test evaluation.

Sweep (training only). Rolling-origin validation over alphabet $a \in \{4, 6, 8\}$, PAA window $w \in \{3, 6, 12\}$ samples, order $k \in \{1, 2, 3\}$, selected at $L=48$ symbols (Table I). Selection: $a=8$, $w=12$ (120 min), $k=2$.

Controls. (i) Label-permutation test, 1000 permutations. (ii) AAFT surrogates generated from the *raw* series, then discretized with the frozen SAX parameters. (iii) Volume confound: per-element normalization plus a mean-level-only baseline on the un-normalized series.

H2 mapping (pre-registered). H_0 : order-0 symbol entropy (bits); $B = \log_2 a$; $C = N$; Q : mean pairwise Jensen–Shannon divergence between per-element order-1 transition distributions (bits); $\alpha = -0.77$, not refitted. N_{meas}^* is the interpolated sequence length at which accuracy crosses the permutation band (bootstrap CI over elements). Consistency holds iff $N_{\text{pred}}^* = H_0 B C Q^{-0.77}$ lies in that CI.

H3. Identical CUSUM (Page–Hinkley as robustness check) on symbolic signals (per-element log-likelihood; sliding Jensen–Shannon divergence of the transition distribution vs. the training reference) and on raw first-order signals (sliding mean, variance), 24 h windows at 6 h stride, thresholds calibrated identically on the training period. Lead time is the raw-signal alarm time minus the symbolic alarm time, bootstrapped over elements. Global seed 42; one command reproduces every result.

IV. RESULTS

H1 (fingerprint): supported. Identification greatly exceeds chance for every N and every sequence length, with permutation $p < 0.001$ throughout (Table II, Fig. 1). At the primary

TABLE II
H1: TEST-A IDENTIFICATION ACCURACY (ORDER-2 MARKOV). EVERY
CELL HAS PERMUTATION $p < 0.001$. CHANCE = $1/N$.

N	chance	$L=12$	$L=24$	$L=48$	$L=96$
10	0.100	0.325	0.475	0.500	0.600
25	0.040	0.315	0.400	0.560	0.680
50	0.020	0.270	0.405	0.510	0.680
100	0.010	0.200	0.320	0.445	0.510

TABLE III
CONTROLS AT $N=50$, $L=48$ (TEST-A). ALL $p < 0.001$.

Classifier / control	accuracy	chance
Markov, order 2 (grammatical)	0.510	0.020
LSTM (sequence model, S1)	0.210	0.020
Mean-level baseline (raw volume)	0.460	0.020
AAFT surrogate (Markov)	0.139 ± 0.034	0.020

operating point ($N=50$, $L=48$) accuracy is 0.51 against a 0.02 chance rate. Accuracy grows with sequence length and degrades gracefully with N (Fig. 2). H_0 is rejected.

Grammar vs. marginal (surrogate control). On AAFT surrogates the Markov classifier falls from 0.51 to 0.139 ± 0.034 (Table III). Most of the identifiability is therefore temporal (grammatical), as claimed; but 0.139 is still $7\times$ chance, so the marginal symbol distribution alone carries a non-trivial share. The grammar claim is thus supported but not exclusive.

Volume confound. The mean-level-only baseline, using nothing but each window’s raw average, reaches 0.46 at $N=50$ and 0.85, 0.60, 0.195 at $N=10, 25, 100$. Volume is thus an independent fingerprint that at small N is *stronger* than the grammatical one. Because the Markov classifier acts on per-element z-normalized data, its 0.51 is volume-independent: the two fingerprints coexist. Honestly stated, “grammatical fingerprint” is one of at least two identifying channels, not the sole explanation of identifiability.

S1 (shallow vs. sequence): shallow suffices. The LSTM reaches only 0.21 at $N=50$, $L=48$ (Table III), well below the Markov model. A sequence model adds nothing here, consistent with modest per-element data; we report this plainly rather than tuning the LSTM further.

H2 (threshold scaling): inconsistent. Accuracy already exceeds the permutation band at the shortest tested length, so the measured threshold is $N^* < 12$ symbols for every N (bootstrap CI degenerate at the boundary, $\geq 995/1000$ resamples left-censored). The prior relation predicts 400–2794 symbols (Table IV), one-to-two orders of magnitude larger. The prediction lies far outside the measured CI: the outcomes are *inconsistent*. We report this as-is; the relation is a pre-registered proposal, not a law, and we did not adjust the mapping to improve agreement.

H3 (drift lead time): supported. During the Christmas regime change (Test-B), symbolic log-likelihood drift alarms first (354 h after 12-13), before the raw mean (408 h) and

TABLE IV
H2: MEASURED VS. PREDICTED DATA-VOLUME THRESHOLD N^* (SYMBOLS). MEASURED IS LEFT-CENSORED AT THE SHORTEST TESTED LENGTH ($L=12$).

N	H_0 (bits)	Q (bits)	N^*_{pred}	N^*_{meas}
10	2.96	0.142	400	< 12
25	2.98	0.217	724	< 12
50	2.98	0.250	1299	< 12
100	2.98	0.228	2794	< 12

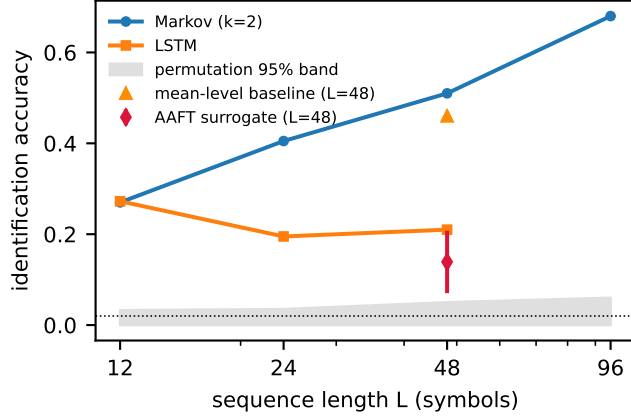


Fig. 1. Test-A accuracy vs. sequence length ($N=50$). Markov exceeds the permutation band at all L ; the LSTM trails; the AAFT surrogate and mean-level points quantify the two controls.

raw variance (468 h); the sliding-JSD signal alarms at 462 h (Fig. 3). The bootstrapped lead of the earliest symbolic over the earliest raw alarm is a median 54 h (95% CI 18–90 h), excluding zero. Grammatical drift precedes first-order drift. As pre-registered context only, Test-B accuracy drops relative to Test-A (e.g. $0.51 \rightarrow 0.16$ at $N=50$), consistent with a regime shift; this exploratory number plays no role in H1.

V. DISCUSSION

Operational upside. A shallow, cheap symbolic model passively identifies the source cell of aggregated telemetry from short windows, and its grammar drifts ~ 2 days before mean/variance during a load regime change. This supports passive element identification and early configuration-/regime-drift detection from data an operator already collects, without per-packet inspection.

Security/privacy downside. The same mechanism is an infrastructure-fingerprinting vector: because identification works on coarse, aggregated, z-normalized activity, an observer who sees only volume envelopes—plausibly behind encryption or aggregation—can still distinguish and track individual elements. Normalization does not protect the element: it merely shifts the discriminative signal from volume to grammar, and both remain exploitable. This argues for treating aggregate telemetry envelopes as identifying data.

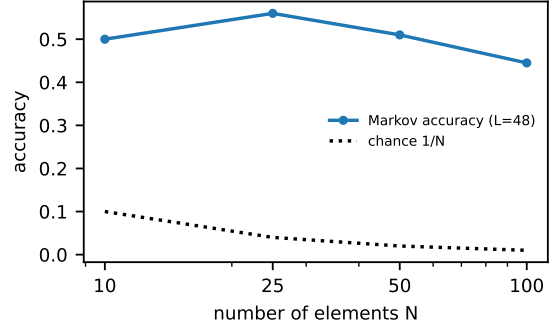


Fig. 2. Test-A accuracy vs. element count N ($L=48$) against the $1/N$ chance line.

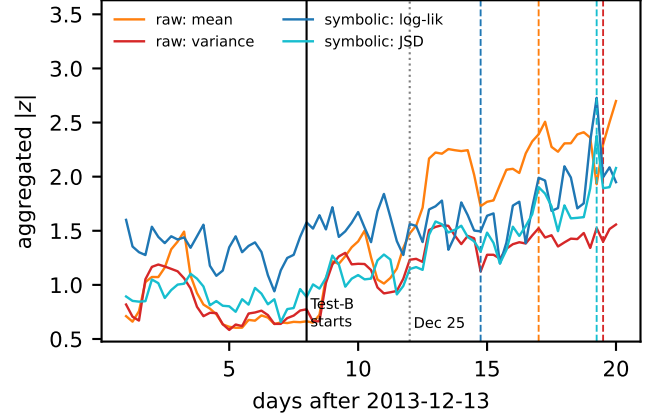


Fig. 3. H3: aggregated deviation of symbolic and raw signals through Test-B. Dashed verticals mark CUSUM alarms; the symbolic log-likelihood alarms earliest.

Substrate independence. Our only quantitative cross-substrate claim is the H2 comparison, and it is *negative*: the Milano threshold does not match the prior quantum/EEG relation (Table IV). We make no claim of a universal law.

VI. LIMITATIONS

Results are from one dataset, one channel, and one city; generality is untested. The strongest limitation is the volume confound: a trivial mean-level classifier rivals or beats the grammatical one at small N , so “fingerprint” must be read as volume-plus-grammar, with grammar isolated only after normalization. The AAFT surrogate leaves a residual $7\times$ -chance marginal component, so the grammatical share, though dominant, is not total. N^* is left-censored below our shortest window, so H2 places only an upper bound; a finer short- L grid could localize it but was not pre-registered and was not added. The prior scaling relation could not be verified in the literature and is treated as a proposal. The LSTM was deliberately small and lightly tuned; a larger model with more per-element data might close the gap to the shallow baseline, though it is not needed for the central claim.

VII. CONCLUSION

Network elements do have a grammatical fingerprint: a shallow Markov model over SAX symbols identifies the source cell far above chance (0.51 vs. 0.02 at $N=50$, $p < 0.001$), the signal is largely temporal (surrogate accuracy 0.139), it is detectable from very short sequences ($N^* < 12$ symbols), and it drifts a median 54h before first-order metrics. The fingerprint is real and volume-independent, but coexists with an equally real volume fingerprint, and its data threshold is inconsistent with the authors’ prior scaling proposal. A sequence model was unnecessary. The honest summary is a supported central claim wrapped in two negative findings—a competing confound and a failed scaling law—reported with equal weight.

DISCLOSURE STATEMENT

Model and framework. All technical content—design, code, analysis, figures, and text—was produced by an autonomous agent (Anthropic Claude, Fable 5) driving the Claude Code / Agent SDK harness. **Human role.** The human supplied the starting prompt and made procedural decisions only: interventions were (1) initial prompt; (2) one *reject* (plan incomplete: H2 mapping and H1/H3 confound), which the agent fixed; (3) one *accept* with a clarifying condition on surrogate generation; (4) one procedural authorization of the email used for the dataset’s mandatory download guestbook. The human wrote no technical content, numbers, citations, or wording. **Pre-registration.** The complete starting prompt, the pre-registered plan, the hyperparameter-freeze commit preceding the single test evaluation, the full commit history, seeds, and the interaction transcript are in the public repository. **Reproducibility.** `python reproduce.py --email <addr>` regenerates every table and figure (global seed 42). **Repository.** <https://github.com/<to-be-inserted-on-submission>>.

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